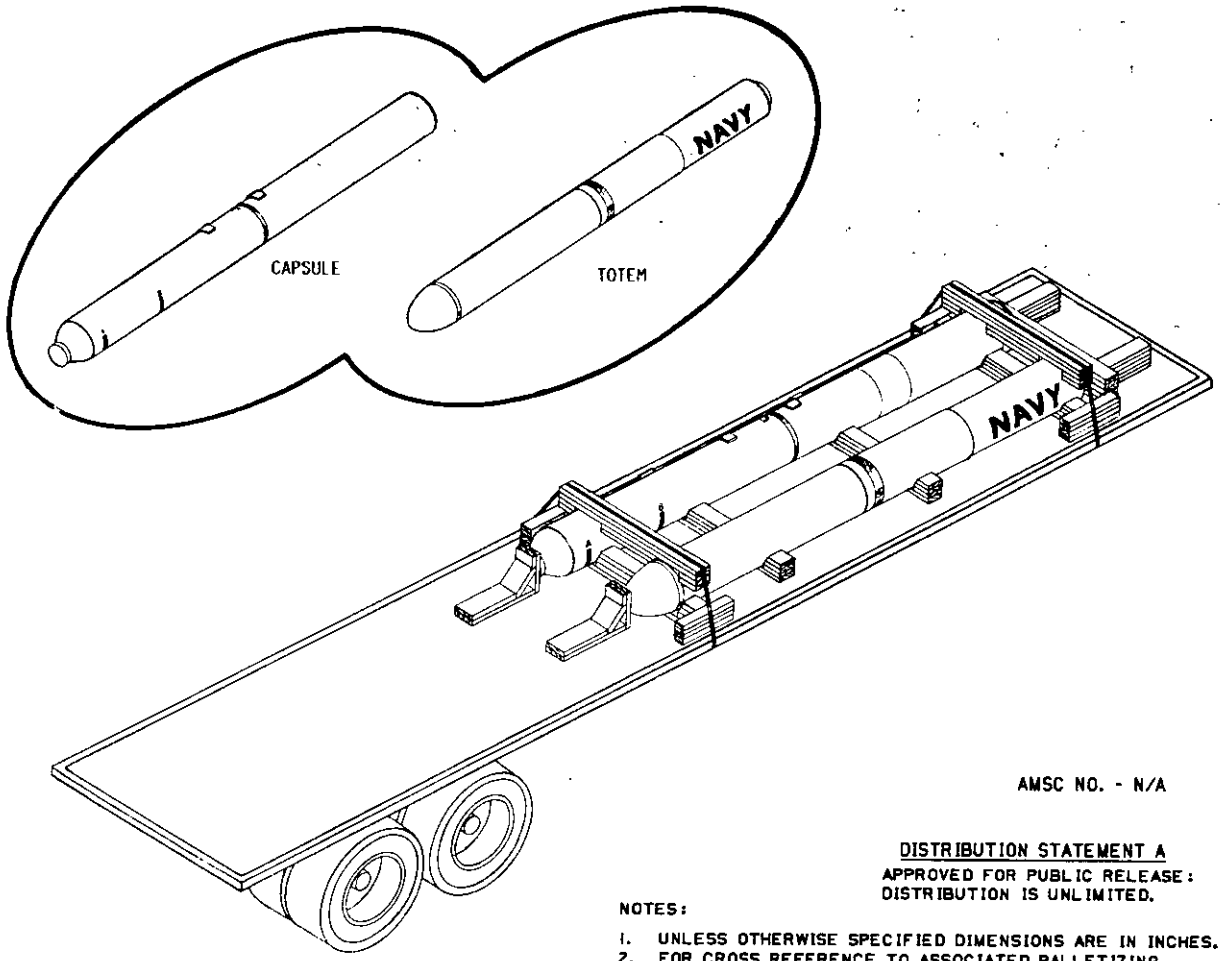
MILITARY STANDARD

MIL-STD-1320-235
(NAVY)
18 NOVEMBER 1988

TRUCKLOADING ENCAPSULATED TOTEM, TOTEM VEHICLE AND EXPENDED CAPSULE

DATA	ENCAP TOTEM	CAPSULE	TOTEM VEHICLE
DODIC/NALC.....			
LENGTH.....	248.2 IN	248.2 IN	243.3 IN
DIAMETER.....	21 IN	21 IN	20.4 IN
GROSS WEIGHT.....	3,500 LBS	900 LBS	2,600 LBS
CUBE.....	50 CU FT	50 CU FT	46 CU FT



AMSC NO. - N/A

DISTRIBUTION STATEMENT A
APPROVED FOR PUBLIC RELEASE: DISTRIBUTION IS UNLIMITED.

NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

REV LTR	REVISION DESCRIPTION	DATE	TDA	SYSCOM
			APPROVAL	

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENT ONLY.

L. Conaldi 11-15-88
SIGNATURE TDA, WPNSTA EARLE DATE

SIGNATURE BY DIRECTION 11 18 88 DATE

ORIGINATOR S Bender 4/15/88 DATE

WPNSTA EARLE, NEW JERSEY

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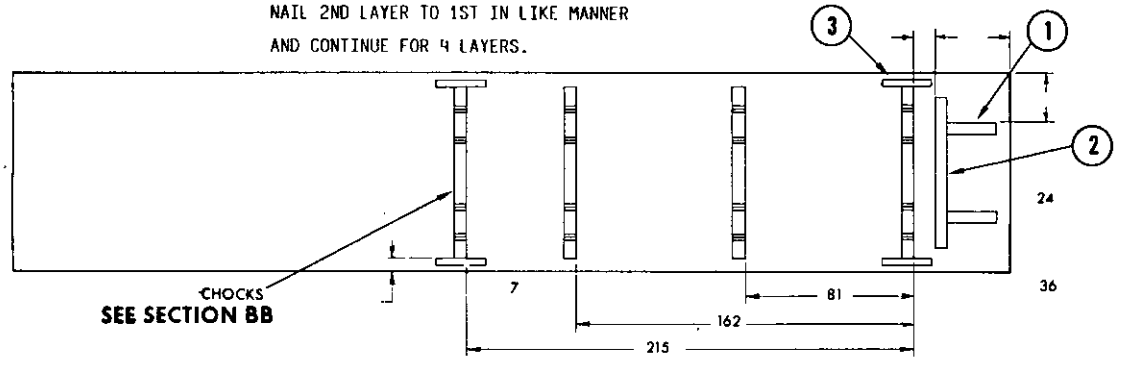
GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE TTL ENCAPSULATED TEST MISSILE, BARE TEST MISSILE OR EMPTY CAPSULE. FOR GENERAL TRUCKLOADING PROCEDURES, REFER TO MIL-STD-1320 (NAVY).
2. THE PROCEDURES DESCRIBED HEREIN ARE INTENDED FOR 28 FT AND LONGER FLATBED TRAILERS. DO NOT USE TRAILERS WITH METAL FLOORS.
3. CHAINS AND STEEL STRAPPING MAY BE USED INTERCHANGEABLY ON A ONE-TO-ONE BASES FOR TIEDOWNS.
4. STEEL STRAPPING USED AS TIEDOWN SHALL BE 2 X .050 AND SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).
5. CHAINS, FITTINGS AND LOAD BINDERS SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY), EXCEPT THAT THE CHAIN/GRAB HOOKS SHALL BE ATTACHED TO THE STAKE POCKETS (NOT AROUND THE RUB RAIL).
6. PROTECTOR BOARDS SHALL BE USED UNDER THE TIEDOWNS. WHEN USING STEEL STRAPPING, THE STRAPS SHALL BE STAPLED TO THE PROTECTOR BOARD USING 2 INCH STRAPPING STAPLES. WHEN USING CHAIN, THE CHAINS SHALL BE SECURED TO THE PROTECTOR BOARD BY DRIVING A 10d NAIL THROUGH THE CHAIN LINK AND CLINCHING THE NAIL OVER THE CHAIN. FIVE STAPLES/NAILS SHALL BE USED FOR EACH TIEDOWN.
7. PRIOR TO LOADING THE TRAILER AND DURING PRELOADING INSPECTION (REQUIRED BY NAVSEA OP 2165 AND REPORTED ON DD FORM 626), THE CHAINS, FITTINGS AND LOAD BINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS, FITTINGS AND LOAD BINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 22 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOAD BINDER.
8. UNLESS OTHERWISE SPECIFIED DUNNAGE LUMBER, STEEL STRAPPING, STRAP SEALS AND NAILS SHALL BE AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).
9. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
10. AFTER THE BLOCKING, BRACING AND TIEDOWNS HAVE BEEN INSPECTED, THE ITEMS SHALL BE COMPLETELY COVERED WITH FIRE RESISTANT, WATERPROOF TARPAULINS.
11. AFTER THE TARPAULINS ARE IN PLACE, ATTACH APPROPRIATE PLACARDS TO BOTH SIDES, FRONT AND BACK OF THE TRAILER (WHEN APPLICABLE) AND ATTACH THE SHIPPING DOCUMENTS TO AN ACCESSIBLE AREA ON THE BACK DECK OF THE TRAILER.

FTL 28 FT AND LONGER TRAILER (FLATBED)

2 X 4 X 24 QUADRUPLED. AFTER ASSEMBLY OF CHOCKS, NAIL ITEM 3 TO TRAILER FLOOR WITH 6 - 16d NAILS. NAIL 2ND LAYER TO 1ST IN LIKE MANNER AND CONTINUE FOR 4 LAYERS.

2 X 6 X 24 QUINTUPLED. NAIL ITEM 1 TO TRAILER FLOOR WITH 3 - 16d NAILS. NAIL 2ND LAYER TO 1ST IN LIKE MANNER AND CONTINUE FOR 5 LAYERS.



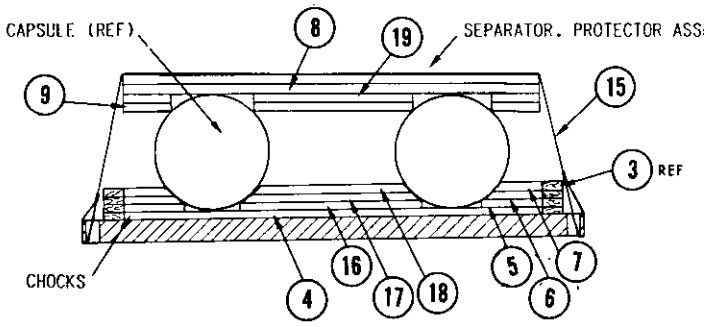
DETAIL A

PREPOSITIONED FLOOR BLOCKING FOR 28 FT OR LONGER TRAILER

2 X 6 X 72 QUINTUPLED. NAIL ITEM 2 TO TRAILER FLOOR WITH 1 - 16d NAIL EVERY 8 INCHES. NAIL SECOND LAYER TO FIRST IN LIKE MANNER AND CONTINUE FOR 5 LAYERS.

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FULL TRUCKLOAD



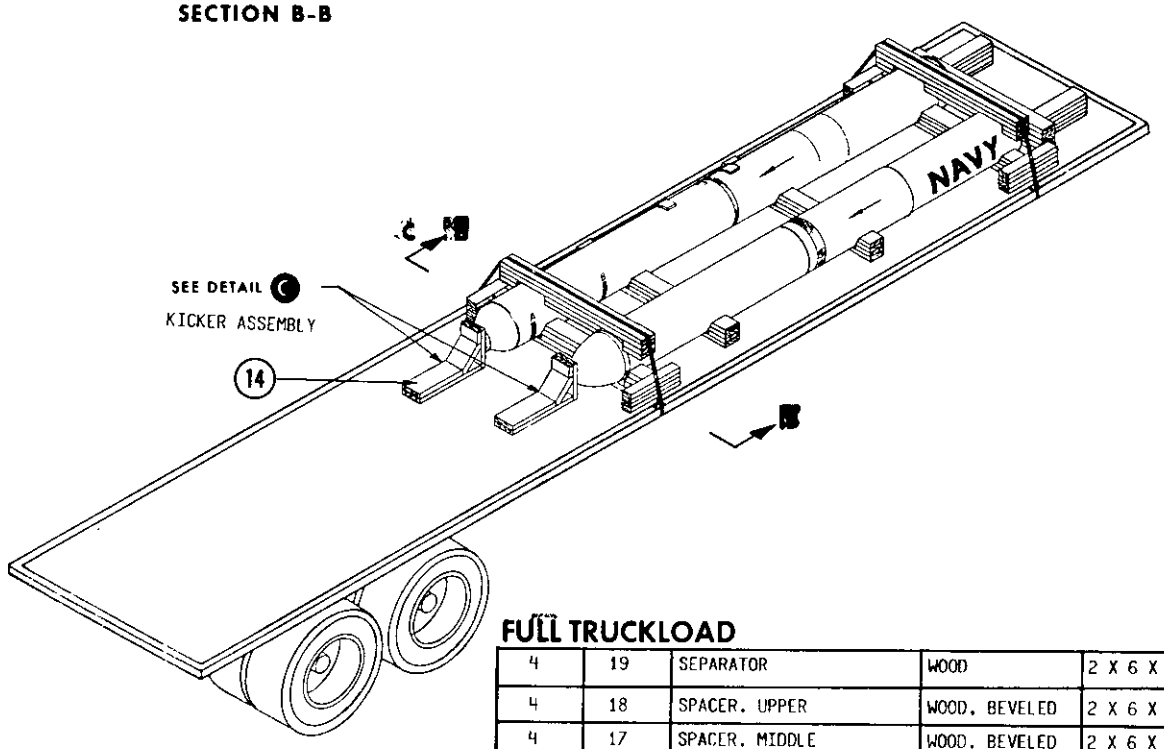
NOTE

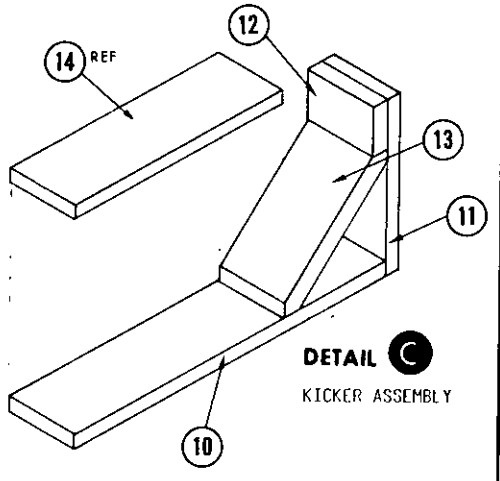
TARPAULINS OMITTED FOR CLARITY.

NOTE

DIRECTION OF ARROW INDICATES FORWARD END OF TOTEM.

SECTION B-B





FULL TRUCKLOAD

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
4	19	SEPARATOR	WOOD	2 X 6 X 33
4	18	SPACER, UPPER	WOOD, BEVELED	2 X 6 X 33
4	17	SPACER, MIDDLE	WOOD, BEVELED	2 X 6 X 36
4	16	SPACER, LOWER	WOOD	2 X 6 X 36
2	15	CHAIN AND LOADER BINDER	SEE NOTE 5	
2	14	TOP CLEAT	WOOD	2 X 8 X 24
2	13	KICKER BOARD	WOOD, BEVELED	2 X 8 X 15
2	12	TOP SUPPORT	WOOD	2 X 8 X 6
2	11	VERTICAL SUPPORT	WOOD	2 X 8 X 18
2	10	BOTTOM CLEAT	WOOD	2 X 8 X 36
8	9	KEEPER BOARD	WOOD	2 X 6 X 9-3/4
4	8	PROTECTOR BOARD	WOOD	2 X 6 X 82
8	7	UPPER CHOCK	WOOD, BEVELED	2 X 6 X 9-3/4
8	6	MIDDLE CHOCK	WOOD, BEVELED	2 X 6 X 11-1/4
8	5	LOWER CHOCK	WOOD	2 X 6 X 11-1/4
4	4	CHOCK BASE	WOOD	2 X 6 X 82
16	3	SLEEPER	WOOD	2 X 4 X 24
5	2	CROSSMEMBER	WOOD	2 X 6 X 72
10	1	BACKUP CLEAT	WOOD	2 X 6 X 24

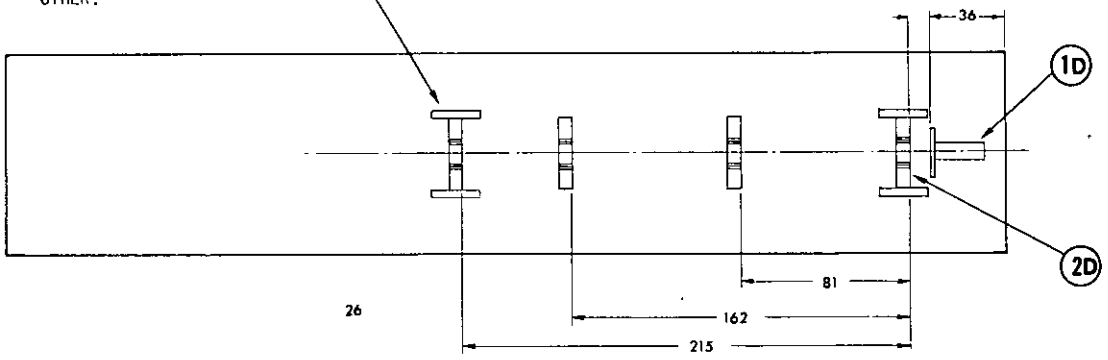
LIST OF MATERIALS

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LESS THAN TRUCKLOAD

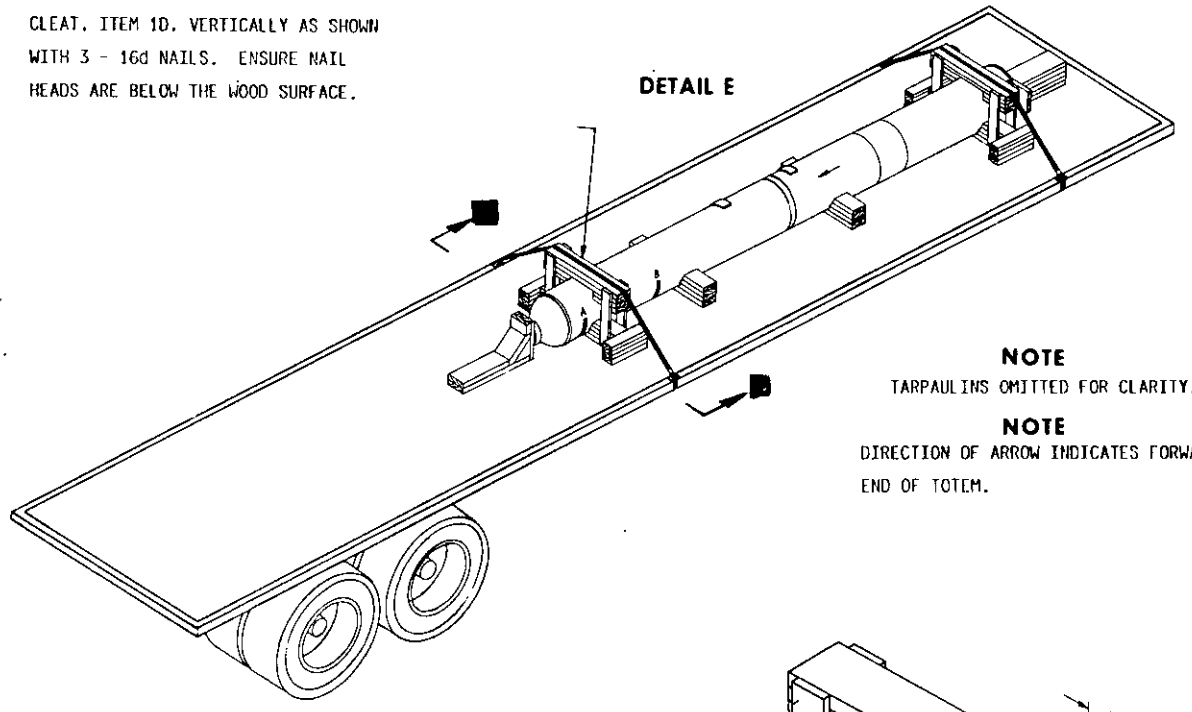
2 X 8 X 24 QUINTUPLED. NAIL ITEM 10 TO TRAILER FLOOR WITH 4 - 16d NAILS. NAIL 2ND LAYER TO 1ST IN LIKE MANNER AND CONTINUE FOR 5 LAYERS.

2 X 4 X 24 QUADRUPLES. SAME AS OTHER.



NAIL CROSSMEMBER, ITEM 2D, TO BACKUP CLEAT, ITEM 1D, VERTICALLY AS SHOWN WITH 3 - 16d NAILS. ENSURE NAIL HEADS ARE BELOW THE WOOD SURFACE.

DETAIL E



NOTE

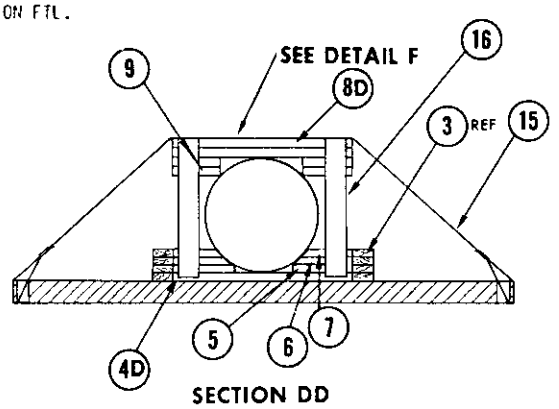
TARPAULINS OMITTED FOR CLARITY.

NOTE

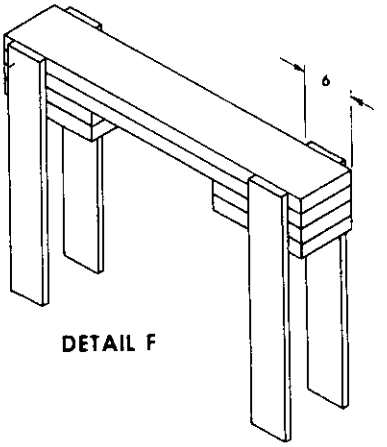
DIRECTION OF ARROW INDICATES FORWARD END OF TOTEM.

NOTE

THE D IN THE ITEM NUMBER REPRESENTS SHORTER VERSIONS OF THE ITEMS CALLED OUT ON FTL.



SECTION DD



DETAIL F

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CHOCK ASSEMBLY INSTRUCTION

1. ASSEMBLE THE CHOCKS ON THE TRUCKS FOLLOWING THESE INSTRUCTIONS. FOR LESS THAN TRUCK (LTL) LOAD USE THE LTL LIST OF MATERIALS AND SECTION D-D.
2. BEFORE STARTING, BEVEL ITEMS 6 AND 7 AND (FOR FULL TRUCK LOADS (FTL) ONLY) AND ITEMS 17 AND 18 AS SHOWN IN SECTION B-B OR SECTION D-D 1-1/2 INCHES BY 45 DEGREES.
3. NAIL CHOCK BASE, ITEM 4, ON THE TRAILER FLOOR IN THE LOCATION INDICATED BY DETAIL A OR DETAIL E WITH 1-16d NAIL EVERY 8 INCHES. REPEAT FOR ALL CHOCK LOCATIONS.
4. NAIL LOWER CHOCK, ITEM 5, ON THE CHOCK BASE WITH 3 - 16d NAILS IN THE LOCATION INDICATED IN SECTION B-B OR SECTION D-D. NEXT NAIL THE MIDDLE CHOCK, ITEM 6, TO THE LOWER CHOCK, AND THE UPPER CHOCK, ITEM 7, TO THE MIDDLE CHOCK IN THE SAME MANNER. REPEAT FOR EACH CHOCK LOCATION.
5. FOR FTL, NAIL THE LOWER SPACER, ITEM 16, TO THE CHOCK BASE WITH 3 - 16d NAILS IN THE LOCATION SHOWN IN SECTION B-B. NAIL THE MIDDLE SPACER, ITEM 17, TO THE LOWER SPACER AND THE UPPER SPACER, ITEM 18, TO THE MIDDLE SPACER IN LIKE MANNER. REPEAT FOR EACH CHOCK LOCATION.

KICKER ASSEMBLY INSTRUCTION

1. PRIOR TO ASSEMBLY, BEVEL 4 EDGES OF KICKER BOARD, ITEM 13, 3/4 BY 45 DEGREES AS SHOWN IN DETAIL C.
2. NAIL TOP SUPPORT, ITEM 12, TO THE END OF THE VERTICAL SUPPORT, ITEM 11, WITH 6 - 8d NAILS. ENSURE NO NAILS PROTRUDE PAST THE FACE OF THE VERTICAL SUPPORT.
3. NAIL THE VERTICAL SUPPORT, ITEM 11, TO THE END OF THE BOTTOM CLEAT, ITEM 10, WITH 3 - 16d NAILS.
4. PLACE THE KICKER BOARD, ITEM 13, AGAINST THE VERTICAL SUPPORT, TOP SUPPORT, AND BOTTOM CLEAT IN THE LOCATION SHOWN ON DETAIL C AT A 45 DEGREE ANGLE. NAIL KICKER BOARD TO THE BOTTOM CLEAT, ITEM 10, WITH 2 - 16d NAILS.
5. AFTER PLACING THE LTL TOTEMS IN THE CHOCK WITH THE AFT END BUTTED AGAINST THE CROSSMEMBER, ITEM 2, PLACE THE KICKER ASSEMBLY AGAINST THE FORWARD END WITH THE VERTICAL SUPPORT, ITEM 11, BUTTED AGAINST THE FORWARD END. NAIL THE ASSEMBLY TO THE TRAILER FLOOR THROUGH THE BOTTOM CLEAT WITH 6 - 16d NAILS. BUTT THE TOP CLEAT, ITEM 14, AGAINST THE KICKER BOARD ABOVE THE BOTTOM CLEAT AND NAIL IN LIKE MANNER. REPEAT PROCEDURE FOR FTL.

PROTECTOR BOARD ASSEMBLY INSTRUCTION

1. NAIL PROTECTOR BOARDS, ITEM 8, TOGETHER WITH 8 - 8d NAILS.
2. NAIL KEEPER BOARD, ITEM 9, TO THE PROTECTOR BOARDS WITH 3 - 16d NAILS IN THE LOCATION SHOWN IN SECTION B-B ((OR SECTION D-D) FOR LTL). NAIL A SECOND KEEPER BOARD TO THE FIRST IN A LIKE MANNER.
3. FOR FTL, NAIL SEPARATOR, ITEM 19, TO THE PROTECTOR BOARD AT THE LOCATION SHOWN IN SECTION B-B WITH 3 - 16d NAILS. NAIL SECOND SEPARATOR BOARD TO FIRST IN LIKE MANNER.
4. REPEAT PROCEDURES FOR EACH PROTECTOR BOARD ASSEMBLY REQUIRED.
5. FOR SINGLE LOAD, NAIL VERTICAL RETAINERS, ITEM 16D, FLUSH WITH THE UPPER PROTECTOR BOARD, ITEM 8, IN THE LOCATION SHOWN IN DETAIL F WITH 4 - 10d NAILS FOR EACH SUPPORT. PLACE PROTECTOR BOARD ASSEMBLY OVER THE CAPSULE AS SHOWN IN SECTION D-D. INSTALL AND TENSION CHAIN AND LOAD BINDERS, ITEM 15, OVER THE PROTECTOR BOARDS PER NOTE 5. NAIL VERTICAL RETAINERS TO THE CHOCK ASSEMBLY WITH 4 - 10d NAILS FOR EACH SUPPORT.

NOTE

VERTICAL RETAINER SHOULD NOT REST ON THE TRAILER FLOOR.

LESS THAN TRUCKLOAD

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
8	16	VERTICAL RETAINER	WOOD	1 X 4 X 25
2	15	CHAIN AND LOAD BINDER	SEE NOTE 5	
1	14	TOP CLEAT	WOOD	2 X 8 X 24
1	13	KICKER BOARD	WOOD, BEVELED	2 X 8 X 15
1	12	TOP SUPPORT	WOOD	2 X 8 X 6
1	11	VERTICAL SUPPORT	WOOD	2 X 8 X 18
1	10	BOTTOM CLEAT	WOOD	2 X 8 X 36
8	9	KEEPER BOARD	WOOD	2 X 6 X 9-5/8
4	8D	PROTECTOR BOARD	WOOD	2 X 6 X 34
8	7	UPPER CHOCK	WOOD, BEVELED	2 X 6 X 9-3/4
8	6	MIDDLE CHOCK	WOOD, BEVELED	2 X 6 X 11-1/4
8	5	LOWER CHOCK	WOOD	2 X 6 X 11-1/4
4	4D	CHOCK BASE	WOOD	2 X 6 X 34-1/4
16	3	SLEEPER	WOOD	2 X 4 X 24
1	2D	CROSSMEMBER	WOOD	2 X 8 X 24
5	1D	BACKUP CLEAT	WOOD	2 X 8 X 24

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